Roll No. ....

TCE–504

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

WATER RESOURCES ENGINEERING—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Draw a neat sketch of an automatic rain gauge (Float type) and describe its working. 10 Marks (CO1)

OR

(b) Station "X" failed to report the rainfall recorded during a storm. With respect to East-West and North-South axes set up at station "X", the coordinates of 4-surrounding gauges, which are nearest to station "X" in respective quadrants are (10, 15), (–8, 5), (–2, –9) and (5, –15) km respectively. Determine the missing rainfall at station "X", if the storm rainfalls at the four surrounding gauges are 73, 89, 68 and 57 mm respectively. 10 Marks (CO1)

2. (a) Explain in brief a method for testing the consistency of rainfall record at a station and necessary adjustment. 10 Marks (CO1)

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OR

(b) A reservoir had an average surface area of 20 km^2 during April, 2022, In that month : 10 Marks (CO1)

Mean rate of inflow = 10 m^3/sec

Mean rate of outflow = 15 m^3/sec

Monthly rainfall = 100 mm

Decrease of storage = 16 million m^3

Assuming seepage losses to be 18 mm

Estimate the evaporation in that month.

3. (a) Discuss the factors affecting runoff from a catchment. 10 Marks (CO2)

OR

(b) Give below are the observed flows from a storm of 3-hr duration on a stream with a drainage area of 40 km^2. 10 Marks (CO2)

time (hr)	Q, m^3/s	BFO, m^3/s
2	50	50
5	47	47
8	75	47
11	120	47
14	225	47
17	290	48
20	270	48
23	145	48

26	110	48
29	90	49
32	80	49
35	70	50
38	60	50
41	55	55
44	51	51
47	50	50

Derive the 3 hr unit hydrograph for the catchments and plot the same.

4. (a) Draw a typical single peaked flood hydrograph and describe the various parts of it. 10 Marks (CO2)

OR

(b) Derive 3-hr synthetic unit hydrograph of basin with the following data with a catchment area of 2500 km^2. 10 Marks (CO2)

Length of main stream = 120 km

Distance from central outlet = 90 km

C_t and C_p of the catchment are assumed to be 1.5 and 0.6 respectively. Use Snyder's method.

5. (a) What is hydrology and hydrologic cycle ? Sketch the block diagram showing "an engineering representation of hydrology". What is the need of studying this subject for Civil Engineers ?

10 Marks (CO1 CO2)

OR

(b) The ordinates of a 3-h UH for a basin are given. Derive the flood hydrograph due to a 3-h storm producing ER of 4 cm. Assume constant base flow of 1 cumec. 10 Marks (CO1 CO2)

Time (hr)	3-hr UHGO
0	0
3	1.5
6	4.5
9	8.6
12	12.0
15	9.4
18	4.6
21	2.3
24	0.8
27	0

90